

SOC Investigation: Brute Force authentication attack detection using Splunk SIEM

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1.Executive Summary:

This investigation simulates a Security Operations Center (SOC) brute force authentication attack scenario using Splunk SIEM. A dataset containing login attempts was analyzed to identify suspicious authentication behavior, including repeated failed login attempts followed by successful access.

The analysis demonstrates how SOC analysts detect credential-based attacks using log correlation, IOC extraction, and timeline reconstruction.

2. Objectives:

The objectives of this SOC investigation were:

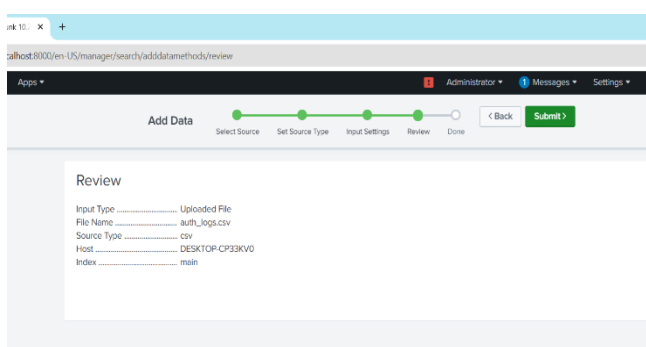
- To analyze authentication logs for suspicious login activity
- To detect brute force attack patterns
- To identify successful logins following failed attempts
- To extract and analyze Indicators of Compromise (IOCs)
- To correlate events into a structured attack timeline
- To map findings to MITRE ATT&CK framework
- To demonstrate SOC workflow using Splunk SIEM

3. Environment & Data Source:

auth_logs.csv - Notepad

```
File Edit Format View Help
time,user,src_ip,country,action,status
2026-05-14 09:00:01,admin,185.220.101.1,Russia,login_attempt,failed
2026-05-14 09:00:05,admin,185.220.101.1,Russia,login_attempt,failed
2026-05-14 09:00:10,admin,185.220.101.1,Russia,login_attempt,failed
2026-05-14 09:00:20,admin,185.220.101.1,Russia,login_attempt,success
2026-05-14 09:01:01,user1,103.25.45.2,China,login_attempt,failed
2026-05-14 09:01:05,user1,103.25.45.2,China,login_attempt,failed
2026-05-14 09:02:10,user2,192.168.1.10,Local,login_attempt,success
2026-05-14 09:03:44,finance,45.77.65.211,Nigeria,login_attempt,failed
2026-05-14 09:03:50,finance,45.77.65.211,Nigeria,login_attempt,failed
2026-05-14 09:04:10,finance,45.77.65.211,Nigeria,login_attempt,success
```

This dataset was created manually using Notepad in CSV format. It stimulates authentication logs for a soc brute force scenario.



The dataset was integrated into Splunk under **index=main** for analysis

COMPONENT	DETAILS
TOOL	Splunk SIEM

DATASET
INDEX
TYPE

CSV authentication logs
Main
Simulated SOC lab

4. Methodology

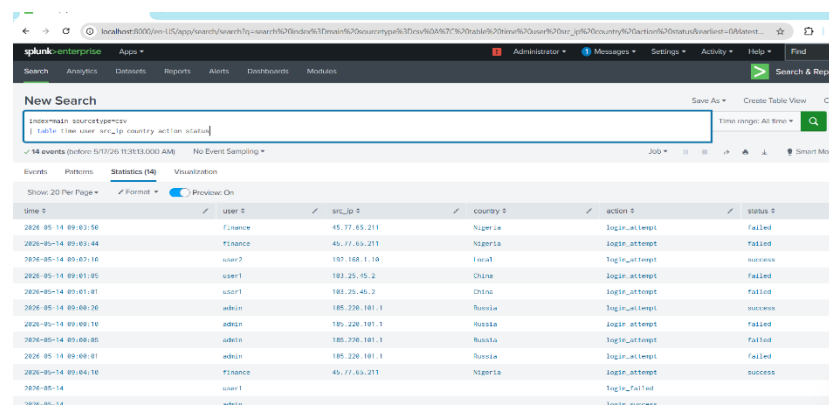
In this investigation, the following steps were performed:

- Data was ingested into Splunk SIEM under index=main
- Authentication logs were analyzed using SPL queries
- Failed login attempts were identified and filtered
- Successful logins were correlated with failed attempts
- Indicators of Compromise (IP addresses and countries) were extracted
- Attack timeline was reconstructed

5. Analysis & Findings

5.1 Field Extraction

The raw authentication logs were parsed and structured using field extraction techniques in Splunk. Key fields such as user, action, IP address, and status were extracted from raw log events for further analysis.

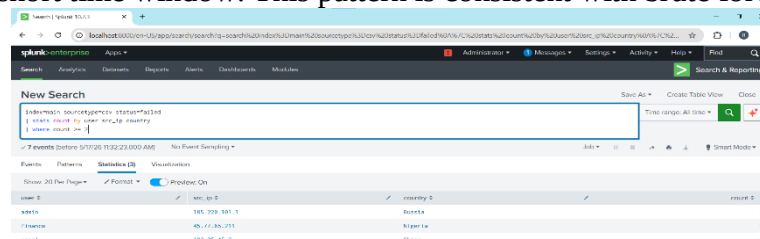


The screenshot shows a Splunk search interface with the query `index=main sourcetype=csv | table time user src_ip country action status`. The results table displays 14 events from 2020-05-14 09:03:58 to 2020-05-14 09:04:16. The columns are time, user, src_ip, country, action, and status. The data shows multiple failed login attempts for users like 'Fluence', 'user2', 'user1', 'usor1', and 'admin' from various IP addresses and countries (Nigeria, Russia, China, Russia).

time	user	src_ip	country	action	status
2020-05-14 09:03:58	Fluence	45.77.45.211	Nigeria	login_attempt	failed
2020-05-14 09:03:58	Fluence	45.77.45.211	Nigeria	login_attempt	failed
2020-05-14 09:03:58	user2	193.25.45.2	Russia	login_attempt	success
2020-05-14 09:03:59	user1	193.25.45.2	China	login_attempt	failed
2020-05-14 09:04:01	usor1	185.226.181.1	Russia	login_attempt	failed
2020-05-14 09:04:06	admin	185.226.181.1	Russia	login_attempt	success
2020-05-14 09:04:10	admin	185.226.181.1	Russia	login_attempt	failed
2020-05-14 09:04:05	admin	185.226.181.1	Russia	login_attempt	failed
2020-05-14 09:04:01	admin	185.226.181.1	Russia	login_attempt	failed
2020-05-14 09:04:10	Fluence	45.77.45.211	Nigeria	login_attempt	success
2020-05-14	user1			login_failed	
2020-05-14	admin			login_success	

5.2 Brute Force Detection

Repeated failed login attempts were identified for specific user accounts within a short time window. This pattern is consistent with brute force authentication attacks.

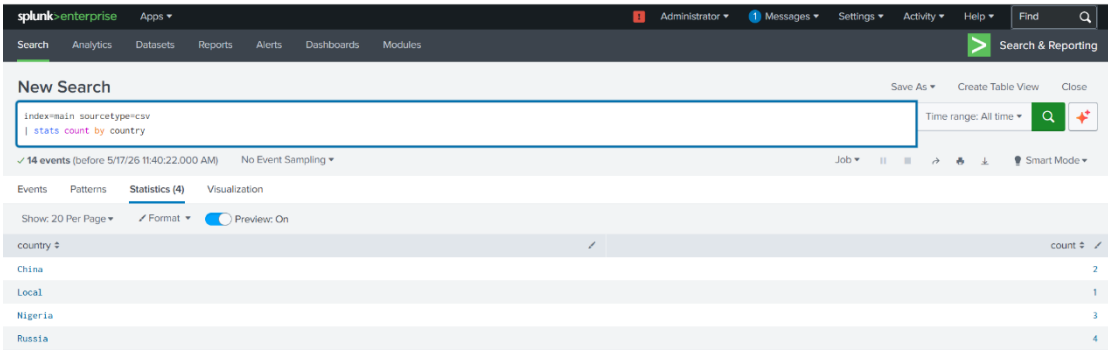


The screenshot shows a Splunk search interface with the query `index=main sourcetype=csv status=failed | stats count by user src_ip country | where count > 1`. The results table displays 7 events, showing counts for users 'admin', 'Fluence', and 'user1' across different IP addresses and countries.

user	src_ip	country	count
admin	185.226.181.1	Russia	3
Fluence	45.77.45.211	Nigeria	2
user1	193.25.45.2	China	2

5.3 Geolocation Analysis

The source IP addresses were analyzed to determine geographic origin. Multiple login attempts originated from suspicious or unexpected locations, indicating potential malicious activity.

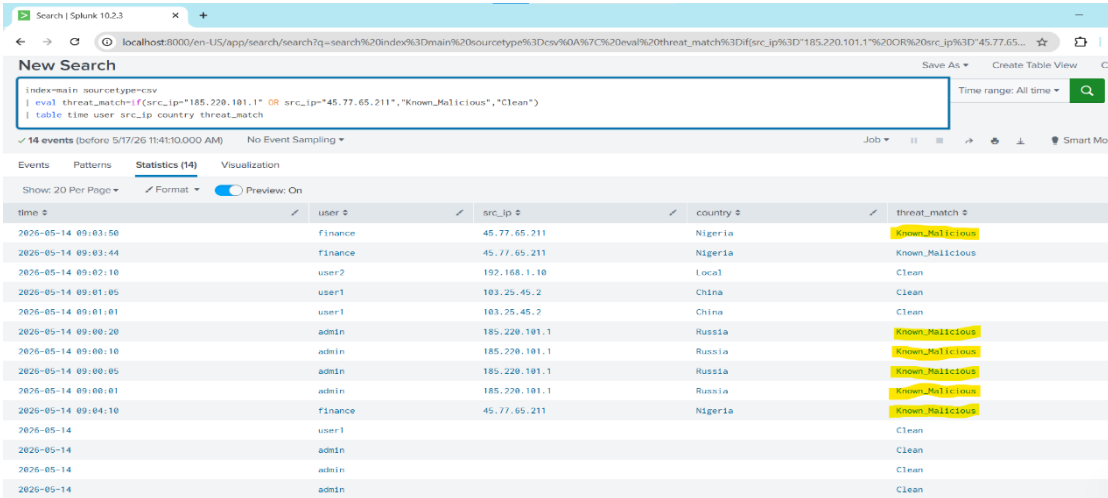


The screenshot shows the Splunk Enterprise interface with a search query: `index=main sourcetype=csv | stats count by country`. The results are displayed in a table with 4 events.

country	count
China	2
Local	1
Nigeria	3
Russia	4

5.4 Threat Intelligence Matching

Identified IP addresses were correlated against known malicious patterns (simulated threat intelligence matching). Some IPs showed behavior consistent with previously observed attack sources.



The screenshot shows the Splunk Enterprise interface with a search query: `index=main sourcetype=csv | eval threat_match=if(src_ip="185.220.101.1" OR src_ip="45.77.65.211", "Known_Malicious", "Clean") | table time user src_ip country threat_match`. The results are displayed in a table with 14 events.

time	user	src_ip	country	threat_match
2026-05-14 09:03:50	Finance	45.77.65.211	Nigeria	Known_Malicious
2026-05-14 09:03:44	Finance	45.77.65.211	Nigeria	Known_Malicious
2026-05-14 09:02:10	user2	192.168.1.10	Local	Clean
2026-05-14 09:01:05	user1	103.25.45.2	China	Clean
2026-05-14 09:01:01	user1	103.25.45.2	China	Clean
2026-05-14 09:00:20	admin	185.220.101.1	Russia	Known_Malicious
2026-05-14 09:00:10	admin	185.220.101.1	Russia	Known_Malicious
2026-05-14 09:00:05	admin	185.220.101.1	Russia	Known_Malicious
2026-05-14 09:00:01	admin	185.220.101.1	Russia	Known_Malicious
2026-05-14 09:04:10	Finance	45.77.65.211	Nigeria	Known_Malicious
2026-05-14	user1			Clean
2026-05-14	admin			Clean
2026-05-14	admin			Clean
2026-05-14	admin			Clean

5.5 IOC (Indicators of Compromise) Analysis

Key indicators of compromise were extracted, including:

- Source IP addresses
- User accounts targeted
- Repeated failed authentication attempts

These indicators suggest possible credential-based attack activity.

New Search

Index=main sourcetype=csv
| stats count by src_ip country status

Time range: All time

✓ 14 events (before 5/17/26 11:33:30.000 AM) No Event Sampling

Events Patterns **Statistics (6)** Visualization

Show: 20 Per Page Format Preview On

src_ip	country	status	count
183.25.45.2	China	failed	2
185.220.101.1	Russia	failed	3
185.220.101.1	Russia	success	1
192.168.1.10	Local	success	1
45.77.65.211	Nigeria	failed	2
45.77.65.211	Nigeria	success	1

5.6 Attack Timeline Reconstruction

Events were sorted chronologically to reconstruct the attack sequence. This allowed visualization of how the authentication attempts progressed over time.

New Search

Index=main sourcetype=csv
| sort _time
| table time user src_ip country status

Time range: All time

✓ 14 events (before 5/17/26 11:44:05.000 AM) No Event Sampling

Events Patterns **Statistics (14)** Visualization

Show: 20 Per Page Format Preview On

time	user	src_ip	country	status
2025-05-14	user1			
2025-05-14	admin			
2025-05-14	admin			
2025-05-14	admin			
2025-05-14 09:00:01	admin	185.220.101.1	Russia	failed
2025-05-14 09:00:05	admin	185.220.101.1	Russia	failed
2025-05-14 09:00:10	admin	185.220.101.1	Russia	failed
2025-05-14 09:00:20	admin	185.220.101.1	Russia	success
2025-05-14 09:01:01	user1	183.25.45.2	China	failed
2025-05-14 09:01:05	user1	183.25.45.2	China	failed
2025-05-14 09:02:10	user2	192.168.1.10	Local	success
2025-05-14 09:03:44	Finance	45.77.65.211	Nigeria	failed
2025-05-14 09:03:50	Finance	45.77.65.211	Nigeria	failed
2025-05-14 09:04:10	Finance	45.77.65.211	Nigeria	success

5.7 Basic Automation

Basic automation logic was applied using Splunk queries to detect patterns of repeated failed login attempts. This helped quickly highlight suspicious users and activity without manual inspection.

The screenshot shows a Splunk Enterprise search interface. The search bar contains a query: `index=main sourcetype=csv | stats count by user, status | eval severity=if(count>3,"High","Medium")`. The results table shows 54 events from 5/17/25 11:45:10.000 AM. The table has columns: user, status, count, and severity. The data rows are as follows:

user	status	count	severity
admin	failed	3	High
admin	success	1	Medium
finance	failed	2	Medium
finance	success	1	Medium
user1	failed	2	Medium
user2	success	1	Medium

6. MITRE ATT&CK Mapping

The observed activity in this investigation was mapped to the MITRE ATT&CK framework to identify attacker techniques used during the simulated authentication attack.

Technique	Id	Description	Evidence in this lab
Brute Force	T1110	Repeated login attempts to guess credentials	Multiple failed login attempts detected for same user account
Valid Accounts	T1078	Use of legitimate credentials after compromise	Successful authentication following repeated failures
IP Scanning	T1595	Attempting access from multiple IP addresses	Multiple source IPs identified in geolocation analysis
Credential Access	TA0006	Attempt to obtain valid login credentials	Observed through repeated authentication failures

7. Conclusion & Recommendations

The analysis confirmed a successful brute force authentication attack targeting multiple administrative and user accounts (pp. 3, 7). The threat actor successfully gained access (status=success) after a wave of rapid, automated failures coming from foreign geolocations (Russia, China, Nigeria) (pp. 3, 5). This pattern highlights a critical vulnerability in exposing authentication endpoints without Multi-Factor Authentication (MFA) or account lockout policies.

Recommendations for Remediation:

- Implement Account Lockout Policies:** Trigger a temporary account lock after 5 consecutive failed logon attempts to prevent automated brute-forcing.

- **Enforce Multi-Factor Authentication (MFA):** Mandate MFA across all corporate and administrative portals to negate the impact of compromised credentials (p. 7).
- **Geofencing / Conditional Access:** Block or flag authentication traffic originating from countries outside of normal operational zones (p. 5).

8. Key Personal Learnings

Building this SOC simulation lab provided invaluable hands-on defensive security skills, specifically:

- **Log Ingestion & Data Structuring:** Learned how raw CSV strings are parsed and processed by a SIEM using index parameters (pp. 3-4).
- **Threat Hunting Patterns:** Mastered the logic of correlating time windows to find successful logons that immediately follow a cluster of failed attempts (pp. 3-4).
- **Threat Intel Integration:** Gained practical understanding of how to cross-reference raw internal event logs against external reputation markers and known malicious entities (p. 5).
- **Framework Alignment:** Learned how to translate technical log data into operational security contexts using standard frameworks like MITRE ATT&CK (p. 7).